

Syllabus:- Binomial Theorem

SECTION - I

Multiple Choice Questions (MCQ)

This section contains **20 multiple choice** questions. Each question has 4 choices (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** is correct.

- The value of $\sum_{r=1}^5 {}^{20}C_{2r-1}$ is :
 (1) 2^{18} (2) 2^{19} (3) 2^{20} (4) $2^{19} - {}^{20}C_{10}$
- If $n \in \mathbb{N}$ and $(1 + x + x^2)^n = a_0 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}$, then $\sum_{r=0}^{2n} a_r =$
 (1) $(-1)^n$ (2) $3^n - 1$ (3) 3^{n-1} (4) 3^n
- If a, b, c be three natural numbers in A.P. and $a + b + c = 21$, then the possible number of values of a, b, c is-
 (1) 15 (2) 14 (3) 13 (4) 16
- The expression $\left(x + \sqrt{x^3 - 1}\right)^5 + \left(x - \sqrt{x^3 - 1}\right)^5$ is a polynomial of degree
 (1) 5 (2) 6 (3) 7 (4) 8
- Which of the following statement is correct ?
 (1) ${}^{50}C_{17}$ is not divisible by 50 (2) $8^7 + 6^7$ is divisible by 896
 (3) If ${}^{49}C_r$ is maximum then $r = 25$ (4) ${}^{2n}C_0^2 + {}^{2n}C_1^2 + \dots + {}^{2n}C_n^2 = 2^{4n-1}$
- The coefficient of x^{20} in the expansion of $(1 + x^2)^{40} \left(x^2 + 2 + \frac{1}{x^2}\right)^{-5}$ is
 (1) ${}^{30}C_{10}$ (2) ${}^{30}C_{25}$ (3) 1 (4) 2
- The coefficient of x^6 in $[(1 + x)^6 + (1 + x)^7 + \dots + (1 + x)^{15}]$ is
 (1) ${}^{16}C_9$ (2) ${}^{16}C_5 - {}^6C_5$ (3) ${}^{16}C_6 - 1$ (4) none of these
- 2^{60} when divided by 7 leaves remainder
 (1) 1 (2) 6 (3) 5 (4) 2

9. Coefficient of x^7 in $(1 - x^4)(1 + x)^9$ is
 (1) -48 (2) 48 (3) -96 (4) 96
10. If in a triangle ABC sines of angles A and B satisfy the equation $4x^2 - 2\sqrt{6}x + 1 = 0$, then $\cos(A - B)$ is equal to –
 (1) 0 (2) $1/2$ (3) $1/\sqrt{2}$ (D) $\sqrt{3}/2$
11. If the coefficients of 5th, 6th, 7th terms in the expansion of $(1 + x)^n$ are in AP then $n =$
 (1) 7 only (2) 14 only (3) 7 or 14 (4) 7 or 21
12. The coefficient of x^{1012} in the expansion of $(1 + x^n + x^{253})^{10}$, (where $n \leq 22$ is any positive integer), is :
 (1) 1 (2) $^{10}C_4$ (3) $4n$ (4) $^{253}C_4$
13. If $(1 + x)(1 + x + x^2)(1 + x + x^2 + x^3) \dots (1 + x + x^2 + \dots + x^n) = a_0 + a_1x + a_2x^2 + \dots + a_mx^m$, then the value of a_1 is
 (1) $m + 1$ (2) $n + 1$ (3) n (4) m
14. If the digits at ten's and hundred's places in $(11)^{2016}$ are x and y respectively, then the ordered pair (x, y) is equal to :
 (1) (1, 8) (2) (1, 6) (3) (6, 1) (4) (8, 1)
15. If angle between tangents drawn to $x^2 + y^2 - 12x - 16y = 0$ at points where curve is cut by $5y = 5x + c$ ($c > 0$) is $\pi/2$, then $c = 10k$ where k equals
 (1) 2 (2) 3 (3) 5 (4) 6
16. Total number of ways in which the letter of the word SUCCESS be arranged so that all s are never together.
 (1) 260 (2) 350 (3) 550 (4) 360
17. If the coefficient of middle term in $(1 + ax)^4$ and that of $(1 - ax)^6$ is same then $a + 1 =$
 (1) 0.5 (2) 2 (3) 0.7 (4) 1
18. If $1 + x^4 + x^5 = \sum_{i=0}^5 a_i(1 + x)^i$, for all x in R , then find value of $a_2 + \frac{19}{2}$
 (1) 5.5 (2) 1.5 (3) 0.5 (4) 0.7
19. Find the coefficient of y^{11} in $\frac{(1+y)^2}{(1-y)^2}$
 (1) 25 (2) 35 (3) 44 (4) 60
20. In a triangle ABC, $\tan \frac{A}{2} = \frac{5}{6}$, $\tan \frac{C}{2} = \frac{2}{5}$, then –
 (1) a, c, b are in A.P. (2) a, b, c are in A.P. (3) b, a, c are in A.P. (4) a, b, c are in G.P.

SECTION-II

(NUMERICAL VALUE TYPE QUESTIONS)

This section contains Five (5) questions. The answer to each question is **NUMERICAL VALUE** with twodigit integer and decimal upto one digit.

21. If $I + F = (\sqrt{2} + 1)^4$ {where I is integer and $f \in [0,1)$ } then find I
22. Find the positive integer just greater than $(1 + 0.0001)^{10000}$
23. Find the number of integral terms in expansion of $(\sqrt{3} + \sqrt[8]{5})^{256}$
24. Find the sum of rational terms in expansion of $(\sqrt{2} + 3^{1/5})^{10}$
25. Find the coefficient of x^{13} in $(1-x)^5 (1+x+x^2+x^3)^4$

ANSWER KEY_DPP-23

1.	(1)	2.	(4)	3.	(3)	4.	(3)	5.	(2)	6.	(2)	7.	(1)
8.	(1)	9.	(1)	10.	(2)	11.	(3)	12.	(2)	13.	(3)	14.	(3)
15.	(4)	16.	(4)	17.	(3)	18.	(1)	19.	(3)	20.	(2)	21.	(33)
22.	(3)	23.	(33)	24.	(41)	25.	(4)						